Beta-Gamma

Q.1) Prove that :

$$\Gamma(n) = \int_0^1 \log(1/y)^{n-1} \, dy \quad (n>0).$$

$\Rightarrow$ let $z = \log(1/y)$

$e^z = 1/y$ $\qquad \left[e^{\log x} = x \right]$

$y = \frac{1}{e^z}$

$y = e^{-z}$

$dy = -e^{-z} dz$

y	0	1
z	∞	0

$$\int_{\infty}^{0} z^{n-1} \cdot -e^{-z} \, dz$$

$$-\int_{\infty}^{0} e^{-z} z^{n-1} \, dz$$

$$\int_{0}^{\infty} e^{-z} z^{n-1} \, dz \qquad \left[-\int_a^b dx = \int_b^a dx \right]$$

$= \Gamma(n)$ proved.

Q.2) Prove that;

$$\beta(p,q) = \int_0^\infty \frac{y^{q-1}}{(1+y)^{p+q}}\, dy$$

let $z = \frac{1}{1+y}$

$$1+y = \frac{1}{z}$$

$$y = \frac{1}{z} - 1$$

$$dy = -\frac{1}{z^2}\, dz$$

y	0	∞
z	1	0

$$\int_1^0 \left(\frac{1}{z} - 1\right)^{q-1} \times (z)^{p+q} \cdot -\frac{1}{z^2}\, dz$$

$$-\int_1^0 \frac{(1-z)^{q-1}}{(z)^{q-1}} \frac{(z)^{p+q}}{z^2}\, dz$$

$$\int_0^1 (1-z)^{q-1} (z)^{p+q-2-q+1}\, dz$$

[$\log(x) \rightarrow$ Negative when value $\rightarrow 0 < x < 1$]

$$\int_0^1 (1-z)^{q-1} (z)^{p-1} \, dz$$

$$\int_0^1 (z)^{p-1} (1-z)^{q-1} \, dz$$

$$= \beta(p, q)$$ proved

Q.3) Express in gamma function;

(a) $$\int_0^1 \frac{dx}{\sqrt{1-x^4}}$$

let $x^2 = \sin\theta$

$$2x \, dx = \cos\theta \, d\theta$$

$$dx = \frac{1}{2} \frac{\cos\theta}{x} \, d\theta$$

$$= \frac{1}{2} \frac{\cos\theta}{\sqrt{\sin\theta}} \, d\theta \qquad [x = \sqrt{\sin\theta}]$$

$$dx = \frac{\cos\theta}{2\sqrt{\sin\theta}} \, d\theta$$

x	0	1
θ	0	$\pi/2$

$$\int_0^{\pi/2} \frac{\cos\theta}{2\sqrt{\sin\theta}} \times \frac{1}{\sqrt{1-\sin^2\theta}} \, d\theta$$

$$\frac{1}{2}\int_0^{\pi/2} \frac{\cos\theta}{\sqrt{\sin\theta}} \times \frac{1}{\sqrt{\cos^2\theta}}\, d\theta$$

$$\frac{1}{2}\int_0^{\pi/2} \frac{\cos\theta}{\sqrt{\sin\theta}} \cdot \frac{1}{\cos\theta}\, d\theta$$

$$\frac{1}{2}\int_0^{\pi/2} (\sin\theta)^{-1/2}\, d\theta$$

$$\frac{1}{2} \times \frac{1}{2}\,\frac{\Gamma\left(\frac{-1/2+1}{2}\right)\Gamma\left(\frac{0+1}{2}\right)}{\Gamma\left(\frac{-1/2+0+2}{2}\right)}$$

$$\frac{1}{4}\,\frac{\Gamma(1/4)\cdot\Gamma(1/2)}{\Gamma(3/4)}$$

$$\frac{1}{4}\,\frac{\sqrt{\pi}\,\Gamma(1/4)}{\Gamma(3/4)} \qquad \left[\Gamma(1/2) = \sqrt{\pi}\right]$$

(b) $$\int_0^{\pi/2} \sqrt{\tan\theta}\, d\theta$$

$$\int_0^{\pi/2} \frac{\sqrt{\sin\theta}}{\sqrt{\cos\theta}}\, d\theta$$

$$\int_0^{\pi/2} (\sin\theta)^{1/2} (\cos\theta)^{-1/2}\, d\theta$$

$$\frac{\frac{1}{2}\,\Gamma\left(\frac{1/2+1}{2}\right)\cdot\Gamma\left(\frac{-1/2+1}{2}\right)}{\Gamma\left(\frac{1/2-1/2+2}{2}\right)}$$

$$\frac{\frac{1}{2}\,\Gamma(3/4)\cdot\Gamma(1/4)}{\Gamma(1)}$$

$$= \frac{\Gamma(3/4)\cdot\Gamma(1/4)}{2}$$

(c) $\int_0^\infty \frac{x^c}{c^x}\,dx$

let $z = c^x$

$\log z = x \log c$

$z = e^{x \log c}$ $\quad [e^{\log x} = x)$

$c^x = e^{x \log c}$

$\int_0^\infty \frac{x^c}{e^{x \log c}}\,dx$

$\int_0^\infty e^{-x \log c} \cdot x^c\,dx$

let $y = x \log c$

$dy = \log c\,dx, \Rightarrow dx = \frac{dy}{\log c}$

$x = \frac{y}{\log c}$

x	0	∞
y	0	∞

$$\int_0^\infty e^{-y} \cdot \left(\frac{y}{\log c}\right)^c \frac{dy}{\log c}$$

$$\frac{1}{\log c}\int_0^\infty e^{-y} \cdot \frac{y^c}{(\log c)^c}\, dy$$

$$\frac{1}{\log c} \times \frac{1}{(\log c)^c} \int_0^\infty e^{-y} \cdot y^c \, dy$$

$$\frac{1}{(\log c)^{c+1}} \Gamma(c+1)$$

(d) $\int_0^\infty a^{-bx^2}\, dx$

let $z = a^{-bx^2}$

$\log z = -bx^2 \log a$

$z = e^{-bx^2 \log a} \Rightarrow a^{-bx^2} = e^{-bx^2 \log a}$

$$\int_0^\infty e^{-bx^2 \log a}\, dx$$

let $y = (b \log a)\, x^2$

$$dy = 2x b\log a \, dx$$

$$dx = \frac{dy}{2x b\log a}$$

$$y = (b\log a)\, x^2$$

$$x^2 = \frac{y}{(b\log a)}$$

$$x = \frac{\sqrt{y}}{\sqrt{(b\log a)}}$$

$$dx = \frac{dy}{2 \times \frac{\sqrt{y}}{\sqrt{b\log a}}\, b\log a}$$

$$dx = \frac{dy}{2\sqrt{y} \times \sqrt{b\log a}}$$

$$\int_0^\infty e^{-y} \cdot \frac{dy}{2\sqrt{y}\sqrt{b\log a}}$$

$$\frac{1}{2\sqrt{b\log a}} \int_0^\infty e^{-y} \cdot (y)^{-1/2}\, dy$$

$$\frac{1}{2\sqrt{b\log a}} \left[\Gamma\left(\tfrac{1}{2}\right)\right]$$

$$= \frac{\sqrt{\pi}}{2\sqrt{b\log a}} \qquad \left[\Gamma\left(\tfrac{1}{2}\right) = \sqrt{\pi}\right]$$

$$\int_0^1 x^5 \left[\log\left(\frac{1}{x}\right)\right]^3 dx$$

let $z = \log\left(\frac{1}{x}\right)$

$$e^z = \frac{1}{x} \qquad \left[e^{\log a} = a\right]$$

$$x = \frac{1}{e^z}$$

$$x = e^{-z} \quad \Rightarrow \quad x^5 = e^{-5z}$$

$$dx = -e^{-z}\, dz$$

x	0	1
z	∞	0

$$\int_\infty^0 e^{-5z} (z)^3 \cdot -e^{-z}\, dz$$

$$-\int_\infty^0 e^{-6z} (z)^3\, dz$$

$$\int_0^\infty e^{-6z} (z)^3\, dz$$

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let $y = 6z$

$z = \frac{y}{6}$

$dz = \frac{1}{6} dy$

$$\int_0^\infty e^{-y} \left(\frac{y}{6}\right)^3 \times \frac{1}{6} dy$$

$$\frac{1}{6^4} \int_0^\infty e^{-y} y^3 dy$$

$$\frac{1}{6^4} \Gamma(4) = \frac{6}{6^4} = \frac{1}{6^3} = \frac{1}{216}$$

Q.4 Evaluate: $\int_0^\infty e^{-ax} x^{m-1} dx$

let $z = ax$

$dz = a\, dx, \Rightarrow dx = \frac{dz}{a}$

$x = \frac{z}{a}$

x	0	∞
z	0	∞

$$\int_0^\infty e^{-z} \cdot \left(\frac{z}{a}\right)^{m-1} \frac{dz}{a}$$

$$\frac{1}{a} \int_0^\infty e^{-z} \cdot \frac{(z)^{m-1}}{a^{m-1}} dz$$

$$\frac{1}{a} \times \frac{1}{a^{m-1}} \int_0^\infty e^{-z} z^{m-1} dz$$

$$\frac{1}{a} \Gamma(m)$$

Q Evaluate ; $\int_0^\infty e^{-ax} x^{m-1} \sin bx \, dx$

let $z = ax$

$dz = a\,dx$

$dx = \frac{dz}{a}$

$x = \frac{z}{a}$

x	0	∞
z	0	∞

$$\int_0^\infty e^{-z} \cdot \left(\frac{z}{a}\right)^{m-1} \sin\left(\frac{bz}{a}\right) \cdot \frac{dz}{a}$$

$$\frac{1}{a} \times \frac{1}{a^{m-1}} \int_0^\infty e^{-z} \cdot z^{m-1} \sin\left(\frac{bz}{a}\right)$$

Q.5 Prove that;

$$\int_0^1 \frac{x^2\,dx}{\sqrt{1-x^4}} \times \int_0^1 \frac{dx}{\sqrt{1+x^4}} = \frac{\pi}{4\sqrt{2}}$$

let $I_1 = \int_0^1 \frac{x^2}{\sqrt{1-x^4}}\,dx$

$$I_2 = \int_0^1 \frac{1}{\sqrt{1+x^2}}\,dx$$

Now, from I_1;

$$I_1 = \int_0^1 \frac{x^2\,dx}{\sqrt{1-x^4}}$$

let $x^2 = \sin\theta$

$2x\,dx = \cos\theta\,d\theta$

$dx = \frac{\cos\theta\,d\theta}{2x}$

$(x = \sqrt{\sin\theta})$

$dx = \frac{\cos\theta}{2\sqrt{\sin\theta}}\,d\theta$

x	0	1
θ	0	$\pi/2$

$$I_1 = \int_0^{\pi/2} \frac{\sin\theta}{\sqrt{1-\sin^2\theta}} \times \frac{\cos\theta}{2\sqrt{\sin\theta}}\,d\theta$$

$$I_1 = \frac{1}{2}\int_0^{\pi/2} \frac{\sin\theta}{\sqrt{\cos^2\theta}} \times \frac{\cos\theta}{\sqrt{\sin\theta}}\, d\theta$$

$$\frac{1}{2}\int_0^{\pi/2} \frac{\sin\theta}{\cos\theta} \times \frac{\cos\theta}{\sqrt{\sin\theta}}\, d\theta$$

$$= \frac{1}{2}\int_0^{\pi/2} \sqrt{\sin\theta}\, d\theta$$

$$= \frac{1}{2}\int_0^{\pi/2} (\sin\theta)^{1/2}\, d\theta$$

$$= \frac{1}{2} \times \frac{\Gamma\left(\frac{1/2+1}{2}\right) \cdot \Gamma\left(\frac{0+1}{2}\right)}{2\,\Gamma\left(\frac{1/2+2}{2}\right)}$$

$$= \frac{1}{2} \times \frac{\Gamma\left(\frac{3}{4}\right) \cdot \Gamma(1/2)}{2 \cdot \Gamma(5/4)}$$

$$= \frac{1}{2}\,\frac{\sqrt{\pi}}{2} \times \frac{\Gamma(3/4)}{\left(\frac{3}{4}-1\right)\Gamma(5/4-1)} \qquad [\Gamma(1/2) = \sqrt{\pi}]$$

$$= \frac{\sqrt{\pi}}{4}\,\frac{\Gamma(3/4)}{\frac{1}{4}\Gamma(1/4)}$$

$$I_1 = \sqrt{\pi}\,\frac{\Gamma(3/4)}{\Gamma(1/4)}$$

Now,

$$I_2 = \int_0^1 \frac{dx}{\sqrt{1+x^4}}$$

let $x^2 = \tan y$

$$2x\,dx = \sec^2 y\,dy$$

$$dx = \frac{\sec^2 y}{2x}\,dy$$

$(x = \sqrt{\tan y})$ $\quad dx = \frac{\sec^2 y}{2\sqrt{\tan y}}\,dy$

x	0	1
y	0	$\pi/4$

$$I_2 = \int_0^{\pi/4} \frac{1}{\sqrt{1+\tan^2 y}} \times \frac{\sec^2 y}{2\sqrt{\tan y}}\,dy$$

$$\int_0^{\pi/4} \frac{1}{\sqrt{\sec^2 y}} \times \frac{\sec^2 y}{2\sqrt{\tan y}}\,dy$$

$$\int_0^{\pi/4} \frac{\sec^2 y}{\sec y\, 2\sqrt{\tan y}}\,dy$$

$$\frac{1}{2}\int_0^{\pi/4} \frac{\sec y}{\sqrt{\tan y}}\, dy$$

$$\frac{1}{2}\int_0^{\pi/4} \frac{1}{\cos y} \times \frac{\sqrt{\cos y}}{\sqrt{\sin y}}\, dy$$

$$\frac{1}{2}\int_0^{\pi/4} \frac{1}{\sqrt{\sin y}\,\sqrt{\cos y}}\, dy$$

$$\frac{1}{2}\int_0^{\pi/4} \frac{1}{\sqrt{\sin y \cos y}}\, dy$$

$$\frac{1}{2}\int_0^{\pi/4} \frac{\sqrt{2}}{\sqrt{2\sin y \cos y}}\, dy$$

$$\frac{1}{\sqrt{2}}\int_0^{\pi/4} \frac{1}{\sqrt{\sin 2y}}\, dy \qquad [\sin 2x = 2\sin x \cos x]$$

$$\frac{1}{\sqrt{2}}\int_0^{\pi/4} (\sin 2y)^{-1/2}\, dy$$

let $p = 2y$

$dp = 2\,dy$

$dy = \frac{dp}{2}$

y	0	$\pi/4$
p	0	$\pi/2$

$$\frac{1}{\sqrt{2}} \int_0^{\pi/2} (\sin p)^{-1/2} \frac{dp}{2}$$ Q.6)

$$\frac{1}{2\sqrt{2}} \int_0^{\pi/2} (\sin p)^{-1/2}\, dp$$

$$\frac{1}{2\sqrt{2}} \times \frac{\Gamma\left(\frac{-1/2+1}{2}\right) \Gamma\left(\frac{0+1}{2}\right)}{2.\Gamma\left(\frac{-1/2+2}{2}\right)}$$

$$\frac{1}{4\sqrt{2}} \frac{\Gamma(1/4)\,\Gamma(1/2)}{\Gamma(3/4)}$$

$$\frac{1}{4\sqrt{2}} \sqrt{\pi}\, \frac{\Gamma(1/4)}{\Gamma(3/4)}$$

$I_1 \times I_2$

$$\frac{\sqrt{\pi}\,\Gamma(3/4)}{\Gamma(1/4)} \times \frac{1}{4\sqrt{2}}\sqrt{\pi}\,\frac{\Gamma(1/4)}{\Gamma(3/4)}$$

$$\frac{\pi}{4\sqrt{2}}$$ proved.

Q.6) Properties;

$$\beta(m, 1/2) = 2^{2m-1}\,\beta(m, m)$$

$$\Gamma(m)\,\Gamma(m+1/2) = \frac{\sqrt{\pi}}{2^{2m-1}}\,\Gamma(2m)$$

Q.) → Evalute $\Gamma(1/4)\Gamma(3/4)$

$$\Gamma(1/4)\,\Gamma\left(1/4 + \frac{1}{2}\right) \qquad \left[\frac{3}{4} = \frac{1}{4} + \frac{1}{2}\right]$$

$$= \frac{\sqrt{\pi}}{2^{2\times 1/4 - 1}}\,\Gamma\left(2 \times \frac{1}{4}\right)$$

$$= \frac{\sqrt{\pi}}{2^{1/2 - 1}}\,\Gamma(1/2)$$

$$= \frac{\sqrt{\pi}}{2^{-1/2}}\,\sqrt{\pi} = \underline{\sqrt{2\pi}}$$

Q.7. Express in terms of gamma functions:

(a) $\int_0^\infty e^{-x^2}\,dx$

let $z = x^2$

$dz = 2x\,dx$

$dx = \frac{dz}{2x}$

$dx = \frac{dz}{2\sqrt{z}}$ $[x = \sqrt{z}]$

x	0	∞
z	0	∞

$$\int_0^\infty e^{-z} \cdot \frac{dz}{2\sqrt{z}}$$

$$\frac{1}{2}\int_0^\infty e^{-z} \cdot (z)^{-1/2}\,dz$$

$$\frac{1}{2}\Gamma(1/2)$$

$$\frac{\sqrt{\pi}}{2}$$

(b) $\int_0^\infty x^6 e^{-3x}\, dx$

let $z = 3x$, $x = \frac{z}{3}$

$dz = 3\,dx$

$dx = \frac{dz}{3}$

x	0	∞
z	0	∞

$$\int_0^\infty \left(\frac{z}{3}\right)^6 \cdot e^{-z} \cdot \frac{dz}{3}$$

$$\frac{1}{3^7} \int_0^\infty e^{-z}\, z^6\, dz$$

$$\frac{1}{3^7}\,\Gamma(7) = \frac{6!}{3^7} = \frac{\overset{2}{\cancel{6}} \times 5 \times 4 \times \cancel{3} \times 2 \times 1}{\cancel{3^7}\; 3^5}$$

$$= \frac{80}{243}$$

(c) $\int_0^\infty \sqrt{x}\, e^{-x^3}\, dx$

let $z = x^3$, $x = (z)^{1/3}$

$dz = 3x^2\, dx$

$dx = \frac{dz}{3x^2}$

$[x^2 = (z)^{2/3}]$

$dx = \frac{dz}{3(z)^{2/3}}$

$\frac{1}{6} - \frac{2}{3}$ $\frac{1-4}{6}$

x	0	∞
z	0	∞

$$\int_0^\infty \sqrt{(z)^{1/3}} \cdot e^{-z} \cdot \frac{dz}{3(z)^{2/3}}$$

$$\frac{1}{3} \int_0^\infty \frac{(z)^{1/6} e^{-z}}{z^{2/3}} \, dz$$

$$\frac{1}{3} \int_0^\infty e^{-z} (z)^{-1/2} \, dz$$

$$\frac{1}{3} \Gamma(1/2)$$

$$\frac{\sqrt{\pi}}{3}$$

Q.8.(a)

(d) $$\int_0^\infty \frac{dx}{x^{p+1} \cdot (x-1)^q} \quad (-p < q < 1)$$

(e) $z = \frac{1}{1-x}$ $\qquad \frac{1}{x-1} = -z$

$1 - x = \frac{1}{z}$, $x - 1 = \frac{1}{z}$

$x = 1 - \frac{1}{z}$

$$dx = +\frac{1}{z^2}\, dz$$

x	0	∞
z	1	0

$$\int_1^0 \frac{(-z)^q}{\left(1-\frac{1}{z}\right)^{p+1}} \cdot \frac{1}{z^2}\, dz$$

$$\int_1^0 \frac{(-z)^q}{\frac{(z-1)^{p+1}}{z^{p+1}}} \times \frac{1}{z^2}\, dz$$

$$\int_1^0 (z-1)^{-p-1} (-z)^q \cdot z^{p-1}\, dz$$

Q.8.(9) show that;

$$\int_0^\infty \frac{x^4}{4^x}\, dx = \frac{\Gamma(5)}{(\log 4)^5}$$

Same as 3(c)

$\boxed{\text{Take } c = 4}$

Q.8.(b) $\int_0^{\pi/2} \sqrt{\cot\theta}\, d\theta = \frac{1}{2}\Gamma(1/4)\Gamma(3/4)$

$$\int_0^{\pi/2} \frac{\sqrt{\cos\theta}}{\sqrt{\sin\theta}}\, d\theta$$

$$\int_0^{\pi/2} (\cos\theta)^{1/2} \times (\sin\theta)^{-1/2}\, d\theta$$

$$\frac{1}{2}\,\frac{\Gamma\left(\frac{1}{2}+\frac{1}{2}\right)\Gamma\left(-\frac{1}{2}+\frac{1}{2}\right)}{\Gamma\left(\frac{\frac{1}{2}-\frac{1}{2}+2}{2}\right)}$$

$$\frac{1}{2}\,\frac{\Gamma(3/4)\Gamma(1/4)}{\Gamma(1)}$$

$$= \frac{1}{2}\Gamma(1/4)\Gamma(3/4)$$

proved.

Ques. $\int_0^{\pi/2} \left(\sqrt{\tan\theta} + \sqrt{\sec\theta}\right) d\theta$

$$\int_0^{\pi/2} \left(\frac{\sqrt{\sin\theta}}{\sqrt{\cos\theta}} + \frac{1}{\sqrt{\cos\theta}}\right) d\theta$$

$$\int_0^{\pi/2} (\sin\theta)^{1/2} (\cos\theta)^{-1/2} \, d\theta + \int_0^{\pi/2} (\cos\theta)^{-1/2} \, d\theta$$

$$\downarrow$$

$$\frac{1}{2} \frac{\Gamma\left(\frac{1/2+1}{2}\right)\Gamma\left(\frac{-1/2+1}{2}\right)}{\Gamma\left(\frac{1/2-1/2+1}{2}\right)} + \frac{1}{2} \frac{\Gamma\left(\frac{0+1}{2}\right)\Gamma\left(\frac{-1/2+1}{2}\right)}{\Gamma\left(\frac{-1/2+2}{2}\right)}$$

$$\frac{1}{2} \frac{\Gamma(3/4)\,\Gamma(1/4)}{\Gamma(1)} + \frac{1}{2} \frac{\Gamma(1/2)\,\Gamma(1/4)}{\Gamma(3/4)}$$

$$\frac{1}{2}\Gamma(1/4)\left[\Gamma(3/4) + \frac{\sqrt{\pi}}{\Gamma(3/4)}\right] \text{ Ans.}$$

(d) $$\underbrace{\int_0^{\pi/2} \sqrt{\sin\theta}\, d\theta}_{I_1} \times \underbrace{\int_0^{\pi/2} \frac{d\theta}{\sqrt{\sin\theta}}}_{I_2} = \pi$$

$$I_1 = \int_0^{\pi/2} (\sin\theta)^{1/2}\, d\theta$$

$$= \frac{1}{2} \frac{\Gamma\left(\frac{1/2+1}{2}\right)\Gamma\left(\frac{0+1}{2}\right)}{\Gamma\left(\frac{1/2+2}{2}\right)}$$

$$= \frac{1}{2} \frac{\Gamma(3/4)\,\Gamma(1/2)}{\Gamma(5/4)}$$

$$= \frac{\sqrt{\pi}}{2} \frac{\Gamma(3/4)}{\left(\frac{5}{4}-1\right)\Gamma(5/4-1)}$$

$$= \frac{\sqrt{\frac{\pi}{2}}}{\frac{1}{4}} \frac{\Gamma(3/4)}{\Gamma(1/4)}$$

$\log_{10} x = a$ $\quad$ $\ln x = a$

$x = 10^a$ $\quad$ $x = e^a$

$$\therefore 2\sqrt{\pi}\,\frac{\Gamma(3/4)}{\Gamma(1/4)}$$

Now, $I_2 = \int_0^{\pi/2} (\sin\theta)^{-1/2}$

$$\frac{1}{2}\,\frac{\Gamma\left(\frac{-1/2+1}{2}\right)\Gamma\left(\frac{0+1}{2}\right)}{\Gamma\left(\frac{-1/2+2}{2}\right)}$$

$$\frac{1}{2}\,\frac{\Gamma(1/4)\,\Gamma(1/2)}{\Gamma(3/4)}$$

$$\frac{\sqrt{\pi}}{2}\,\frac{\Gamma(1/4)}{\Gamma(3/4)}$$

$I_1 \times I_2$

$$2\sqrt{\pi}\,\frac{\Gamma(3/4)}{\Gamma(1/4)} \times \frac{\sqrt{\pi}}{2}\,\frac{\Gamma(1/4)}{\Gamma(3/4)}$$

$= \pi$ $\quad$ Ans proved

~~(e)~~ $\cancel{\int_0^\infty e^{-x^4}\,dx \times \int_0^\infty x^2 e^{-x^4}\,dx = \frac{\pi}{8\sqrt{2}}}$

(e) $\underbrace{\int_0^\infty e^{-x^4}\,dx}_{I_1} \times \underbrace{\int_0^\infty x^2 e^{-x^4}\,dx}_{I_2} = \frac{\pi}{8\sqrt{2}}$

$$I_1 = \int_0^\infty e^{-x^4}\,dx$$

let $z = x^4$

$dz = 4x^3\,dx$

$dx = \frac{dz}{4x^3}$

x	0	∞
z	0	∞

$dx = \frac{dz}{4(z)^{3/4}}$ $\quad [x = (z)^{1/4}]$

$$= \int_0^\infty e^{-z} \cdot \frac{dz}{4z^{3/4}}$$

$\frac{2-3}{4}$ $n = 1 - \frac{3}{4} = 1/4$

$$\frac{1}{4}\int_0^\infty e^{-z} \cdot (z)^{-3/4}\, dz$$

$$I_1 = \frac{1}{4}\,\Gamma(1/4)$$

$$I_2 = \int_0^\infty x^2 e^{-x^4}\, dx$$

let $y = x^4$

$dy = 4x^3\, dx$

$dx = \frac{dy}{4(y)^{3/4}}$

$$\int_0^\infty (y)^{1/2} \cdot e^{-y} \cdot \frac{dy}{4(y)^{3/4}}$$

$$\frac{1}{4}\int_0^\infty e^{-y} \cdot (y)^{-1/4}\, dy$$

$$I_2 = \frac{1}{4}\,\Gamma(3/4)$$

$$\Rightarrow I_1 \times I_2 = \frac{1}{4}\Gamma(1/4) \cdot \frac{1}{4}\Gamma(3/4)$$

$$\frac{1}{16}\,\Gamma(1/4)\,\Gamma(3/4)$$

$$\frac{1}{16}\cdot\left[\Gamma(1/4)\,\Gamma(1/4+1/2)\right]$$

$$\frac{1}{16}\left[\frac{\sqrt{\pi}}{2^{2\times 1/4-1}}\,\Gamma(2\times 1/4)\right]$$ [using property → Q.6].

$$\frac{1}{16}\left(\frac{\sqrt{\pi}}{2^{-1/2}}\,\Gamma(1/2)\right.$$

$$\frac{\pi}{16}\times\sqrt{2}=\frac{\pi}{8\sqrt{2}}$$ proved

d) $\int_0^1 \sqrt[3]{x \log\left(\frac{1}{x}\right)}\, dx = \left(\frac{3}{4}\right)^{4/3} \Gamma(4/3)$

$$\int_0^1 \left(x \log(1/x)\right)^{1/3} dx$$

let $z = \log(1/x)$

$$e^z = \frac{1}{x}$$

$$x = \frac{1}{e^z}$$

$$x = e^{-z}$$

$$dx = -e^{-z}\, dz$$

x	0	1
z	∞	0

$$\int_\infty^0 \left(e^{-z} \cdot z\right)^3 \cdot -e^{-z}\, dz$$

$$-\int_\infty^0 e^{-3z} \cdot e^{-z} \cdot z^3\, dz$$

$$\int_0^\infty e^{-4z} \cdot z^3\, dz$$

let $p = 4z$

$\overline{25^6}$

$$z = \frac{p}{4}$$

$$dz = \frac{1}{4} dp$$

$$\int_0^\infty e^{-p} \cdot \left(\frac{p}{4}\right)^3 \cdot \frac{1}{4} dp$$

$$\frac{1}{4^4} \int_0^\infty e^{-p} p^3 dp$$

$$\frac{1}{256} \Gamma(4)$$

$$\frac{3!}{256} = \frac{3}{128}$$

Q.9) Given $\int_0^\infty \frac{x^{n-1}}{1+x}\,dx = \frac{\pi}{\sin n\pi}$. Show that

$$\Gamma(n)\,\Gamma(1-n) = \frac{\pi}{\sin(n\pi)}$$, Hence

evaluate $\int_0^\infty \frac{dy}{1+y^4}$

let $z = \frac{1}{1+x}$ let

$1+x = \frac{1}{z}$

$x = \frac{1}{z} - 1$

$dx = -\frac{1}{z^2}\,dz$

x	0	∞
z	1	0

$$\int_1^0 \left(\frac{1}{z}-1\right)^{n-1} \cdot z \cdot + \frac{1}{z^2}\,dz$$

$$\int_1^0 \frac{(1-z)^{n-1}}{z^{n-1}} \times \frac{1}{z}\,dz$$

$$-\int_0^1 z^n (1-z)^{n-1}\,dz$$

$$-\beta(n+1, n)$$

$$\cancel{-\beta(n+1,n) = \frac{\Gamma(n+1)\cdot\Gamma(n)}{\Gamma(n+1+n)}}$$

$$\cancel{= \frac{\Gamma(n)\,\Gamma(n+1)}{\Gamma(2n+1)}}$$

~~let $z = 1+x$~~

~~let~~

let $z = \dfrac{1}{1+x}$

$$1+x = \frac{1}{z}$$

$$x = \frac{1}{z} - 1$$

$$dx = -\frac{1}{z^2}\,dz$$

x	0	∞
z	1	0

$$\int_1^0 \left(\frac{1}{z} - 1\right)^{n-1} \cdot z \cdot -\frac{1}{z^2}\,dz$$

$$= \int_1^0 \frac{(1-z)^{n-1}}{z^{n-1}} \frac{1}{z} \, dz$$

$$\int_0^1 z^{-n} (1-z)^{n-1} \, dz$$

$$\left[\beta(m,n) = \int_0^1 x^{m-1} (1-x)^{n-1} \, dx \right]$$

$\beta(1-n, n)$

$$\beta(1-n, n) = \frac{\Gamma(1-n)\Gamma(n)}{\Gamma(1-n+n)}$$

$$= \frac{\Gamma(n)\Gamma(1-n)}{\Gamma(1)}$$

$$= \Gamma(n)\Gamma(1-n) \text{ proved.}$$

Now; $\displaystyle\int_0^\infty \frac{dy}{1+y^4}$

let $y^2 = \tan\theta$

$$2y \, dy = \sec^2\theta \, d\theta$$

$$dy = \frac{\sec^2\theta}{2y} \, d\theta$$

$$dy = \frac{\sec^2\theta}{2\sqrt{\tan\theta}}\, d\theta$$

y	0	∞
θ	0	$\pi/2$

$$\int_0^{\pi/2} \frac{1}{1+\tan^2\theta} \times \frac{\sec^2\theta}{2\sqrt{\tan\theta}}\, d\theta$$

$$\frac{1}{2}\int_0^{\pi/2} \frac{1}{\cancel{\sec^2\theta}} \times \frac{\cancel{\sec^2\theta}}{\sqrt{\tan\theta}}\, d\theta$$

$$\frac{1}{2}\int_0^{\pi/2} \sqrt{\cot\theta}\, d\theta$$

$$\frac{1}{2}\int_0^{\pi/2} (\cos\theta)^{+1/2} \, (\sin\theta)^{-1/2}$$

$$\frac{1}{2} \times \frac{\Gamma\left(\frac{1/2+1}{2}\right)\Gamma\left(\frac{-1/2+1}{2}\right)}{2\,\Gamma\left(\frac{-1/2+1/2+2}{2}\right)}$$

$$\frac{1}{4} \quad \frac{\Gamma(3/4)\,\Gamma(1/4)}{\Gamma(1)}$$

$$\frac{1}{4}\,\Gamma(3/4)\cdot\Gamma(1/4) = \frac{1}{4}\,\Gamma(1/4)\,\Gamma(1/4+1/2)$$

$$\frac{1}{4}\left[\frac{\sqrt{\pi}}{2^{2\times 1/4-1}}\,\Gamma(2\times 1/4)\right.$$

$$\frac{1}{4}\left[\frac{\sqrt{\pi}}{2^{-1/2}}\,\Gamma(1/2)\right]$$

$$\frac{\sqrt{2\pi}}{4} = \frac{\pi}{2\sqrt{2}}$$

Q10) Prove that;

(a) $$\int_0^1 \frac{x\,dx}{\sqrt{1-x^5}} = \frac{1}{5}\,\beta\left(\frac{2}{5},\frac{1}{2}\right)$$

let $z = 1 - x^5 \Rightarrow x^5 = 1 - z$

$x = (1-z)^{1/5}$

$$dz = -5x^4\,dx$$

$$dx = \frac{-dz}{5x^4}$$

$$dx = \frac{-dz}{5(1-z)^{4/5}}$$

x	0	1
z	1	0

(b)

$$\int_1^0 \frac{(1-z)^{1/5}}{\sqrt{z}} \frac{-dz}{5(1-z)^{4/5}}$$

$$-\frac{1}{5}\int_1^0 (z)^{-1/2} \frac{(1-z)^{1/5}}{(1-z)^{4/5}}\, dz$$

$$\frac{1}{5}\int_0^1 (z)^{-1/2} (1-z)^{-3/5}\, dz$$

$$\frac{1}{5}\beta\left(\frac{1}{2}, \frac{2}{5}\right)$$ Ans.

(b) $\int_0^1 \frac{dx}{\sqrt{(1+x^4)}} = \frac{1}{4\sqrt{2}} \beta\left(\frac{1}{4}, \frac{1}{2}\right)$

let $x^2 = \tan\theta$

$2x \frac{dx}{d\theta} = \sec^2\theta \, d\theta$

$dx = \frac{\sec^2\theta \, d\theta}{2\sqrt{\tan\theta}}$

x	0	1
θ	0	$\pi/4$

$$\int_0^{\pi/4} \frac{1}{\left(\sqrt{1+\tan^2\theta}\right)} \frac{\sec^2\theta}{2\sqrt{\tan\theta}} \, d\theta$$

$$\frac{1}{2} \int_0^{\pi/4} \frac{\sec^2\theta}{\sqrt{\sec^2\theta}} \frac{1}{\sqrt{\tan\theta}} \, d\theta$$

$$\frac{1}{2} \int_0^{\pi/4} \frac{\sec^2\theta}{\sqrt{\tan\theta}} \, d\theta$$

$$\frac{1}{2} \int_0^{\pi/4} \frac{1}{\cos\theta} \times \frac{\sqrt{\cos\theta}}{\sqrt{\sin\theta}} \, d\theta.$$

$$\frac{1}{2}\int_0^{\pi/4} \frac{1}{\sqrt{\sin\theta\cos\theta}}\, d\theta$$

$$\frac{1}{2}\int_0^{\pi/4} \frac{\sqrt{2}}{\sqrt{2\sin\theta\cos\theta}}\, d\theta$$

$$\frac{1}{\sqrt{2}}\int_0^{\pi/4} \frac{1}{(\sin 2\theta)^{1/2}}\, d\theta$$

$$\frac{1}{\sqrt{2}}\int_0^{\pi/4} (\sin 2\theta)^{-1/2}\, d\theta$$

let $z = 2\theta$

$\theta = \frac{z}{2}$

$d\theta = \frac{1}{2}\, dz$

2θ	0	$\pi/4$
z	0	$\pi/2$

$$\frac{1}{\sqrt{2}}\int_0^{\pi/2} (\sin z)^{-1/2}\, \frac{1}{2}\, dz$$

$\alpha^{1/2}$ $\frac{1}{2}x$

$2m-1 = -\frac{1}{2}$ | $2n-1=0$

$2m = 1/2$ | $n=\frac{1}{2}$

$m = 1/4$

$$\frac{1}{2\sqrt{2}} \int_0^{\pi/2} (\sin z)^{-1/2} \, dz$$

$$\left[\frac{\beta(m,n)}{2} = \sin^{2m-1}\theta \cos^{2n-1}\theta \right]$$

$$\frac{1}{2\sqrt{2}} \frac{\beta(1/4, 1/2)}{2}$$

$$= \frac{1}{4\sqrt{2}} \beta(1/4, 1/2) \text{ proved.}$$

(c) $$\int_0^1 x^3 (1-\sqrt{x})^5 \, dx = 2\beta(8,6).$$

let $z = 1 - \sqrt{x}$

$x = (1-z)^2$

$dx = -2(1-z)\,dz$

x	0	1
z	1	0

$$\int_1^0 (1-z)^6 \cdot z^5 \left(-2(1-z)\right) dz$$

$$-2\int_1^0 z^5 (1-z)^7 \, dz$$

$$2\int_0^1 z^5 (1-z)^7 \, dz$$

$$\left(B(m,n) = x^{m-1} \cdot (1-x)^{n-1} \right)$$

$$2\,\beta(6,8)$$

proved.

Q.11. proved $\Rightarrow$ $\displaystyle\int_0^\infty \frac{x^{10} - x^{18}}{(1+x)^{30}} \, dx = 0$

let $z = \dfrac{1}{1+x}$

$1+x = \dfrac{1}{z}$

Q.11. $\displaystyle\int_0^\infty \frac{x^{10} - x^{18}}{(1+x)^{30}}\, dx = 0$

$$\underbrace{\int_0^\infty \frac{x^{10}}{(1+x)^{30}}\, dx}_{I_1} - \underbrace{\int_0^\infty \frac{x^{18}}{(1+x)^{30}}\, dx}_{I_2}$$

I_1 = let $z = \dfrac{1}{1+x}$

$$1+x = \frac{1}{z}$$

$$x = \frac{1}{z} - 1$$

$$dx = -\frac{1}{z^2}\, dz$$

x	0	∞
z	1	0

$$\int_1^0 \left(\frac{1}{z} - 1\right)^{10} \times z^{30} \cdot -\frac{1}{z^2}\, dz$$

$$-\int_1^0 \frac{(1-z)^{10}}{z^{10}} \times \frac{z^{30}}{z^2}\, dz$$

$$\int_0^1 z^{18} (1-z)^{10}\, dz$$

$I_1 = \beta(\underline{19, 11})$

$I_2 \Rightarrow$ let $p = \frac{1}{1+x}$

$$1+x = \frac{1}{p}$$

$$x = \frac{1}{p} - 1$$

$$dx = -\frac{1}{p^2}\, dp$$

9	0	2
2	1	0

$$\int_1^0 \left(\frac{1}{p} - 1\right)^{18} \times p^{30} \times \frac{-1}{p^2}\, dp$$

$$-\int_1^0 \frac{(1-p)^{18}}{p^{18}}\, p^{30} \times \frac{1}{p^2}\, dp$$

$$\int_0^1 p^{10} (1-p)^{18}\, dp$$

$I_2 = \beta(11, 19)$

$I_1 - I_2$

$\beta(19, 11) - \beta(11, 19)$ $\quad [\beta(m,n) = \beta(n,m)]$

$= 0$

Q.12 Evaluate: $\int_0^1 x(\log x)^3 \, dx$ and

$$\int_0^1 x^m (\log x)^n \, dx = \frac{(-1)^n \, n!}{(m+1)^{n+1}}$$

$$\int_0^1 x (\log x)^3 \, dx$$

let $-z = \log x$

$e^{-z} = x$

$dx = -e^{-z} dz$

x	0	1
z	∞	0

$$\int_{\infty}^{0} e^{-z} (-z)^3 \cdot -e^{-z} \, dz$$

$$\int_{\infty}^{0} e^{-2z} \, z^3 \, dz$$

let $p = 2z$

$z = \frac{p}{2}$

$dz = \frac{1}{2} dp$

z	0	∞
p	0	∞

$$\int_{\infty}^{0} e^{-p} \left(\frac{p}{2}\right)^3 \times \frac{1}{2} \, dp$$

$$\frac{1}{2^4} \int_{\infty}^{0} e^{-p} p^3 \, dp$$

$$-\frac{1}{2^4} \int_{0}^{\infty} e^{-p} p^3 \, dp$$

$$-\frac{1}{2^4} \Gamma(4) = -\frac{3!}{2^4} = -\frac{3}{8}$$

$$\int_0^1 x^m (\log x)^n \, dx$$

let $-z = \log x$

$$e^{-z} = x$$

$$dx = -e^{-z} dz$$

x	0	1
z	∞	0

$$\int_\infty^0 e^{-z} (-z)^n \, -e^{-z} \, dz$$

$$-\int_\infty^0 e^{-2z} (-z)^n \, dz$$

let $p = 2z$

$$z = \frac{p}{2}$$

$$dz = \frac{1}{2} dp$$

z	0	∞

$$\cancel{\int_0^\infty e^{-r} \left(-\frac{r}{2}\right)^n dz}$$

$$\int_\infty^0 e^{-zm} (-z)^n - e^{-z} \, dz$$

$$-\int_\infty^0 e^{-z(m+1)} (-z)^n \, dz$$

$$\int_0^\infty e^{-z(m+1)} (-z)^n \, dz$$

$$\int_0^\infty e^{-z(m+1)} (z)^n (-1)^n \, dz$$

$$(-1)^n \int_0^\infty e^{-z(m+1)} z^n \, dz$$

let $p = z(m+1)$

$$z = \frac{p}{m+1}$$

$$dz = \frac{1}{m+1} dp$$

z	0	∞
p	0	∞

$$(-1)^n \int_0^\infty e^{-p} \left(\frac{p}{m+1}\right)^n \frac{1}{m+1}\, dp$$

$$\frac{(-1)^n}{(m+1)^n} \times \frac{1}{m+1} \int_0^\infty e^{-p} p^n \, dp$$

$$\frac{(-1)^n}{(m+1)^{n+1}} \cdot \Gamma(n+1)$$

$$\frac{(-1)^n}{(m+1)^{n+1}} \cdot (n)! \qquad [\Gamma(n) = (n-1)!]$$

proved.

Q13. Show that $\int_0^1 y^{q-1} \left(\log \frac{1}{y}\right)^{p-1} dy = \frac{\Gamma(p)}{q^p}$ where $p>0$, $q>0$.

$(-z)(q-1) - z$
$-z(q-1+1)$

let $z = \log\left(\frac{1}{y}\right)$

$$e^{z} = \frac{1}{y}$$

$$y = \frac{1}{e^{z}}$$

$$y = e^{-z}$$

$$dy = -e^{-z}\, dz$$

y	0	1
z	∞	0

$$\int_{\infty}^{0} (e^{-z})^{q-1} \, (z)^{p-1} \cdot -e^{-z}\, dz$$

$$-\int_{\infty}^{0} e^{-z(q-1)-z} z^{p-1}\, dz$$

$$\int_{0}^{\infty} e^{-z(q)} z^{p-1}\, dz$$

let $a = zq$

$$z = \frac{a}{q}$$

$$dz = \frac{1}{q}\, da$$

z	0	∞
a	0	∞

$$\int_0^\infty e^{-q} \left(\frac{q}{q}\right)^{p-1} \frac{1}{q}\, dq$$

$$\frac{1}{q^{p-1}} \times \frac{1}{q} \int_0^\infty e^{-q}\, q^{p-1}\, dq$$

$$\frac{1}{q^p}\, \Gamma(p) \quad \text{proved.}$$

Q.(4.) $\int_0^1 x^m (1-x^n)^p\, dx$ in gama function.

$\Rightarrow$ let $z = 1 - x^n$

$x^n = 1 - z$

$n x^{n-1}\, dx = -dz$

$$dx = -\frac{1}{n} \frac{dz}{x^{n-1}}$$

$$\left[x = (1-z)^{1/n} \right]$$

$$dx = -\frac{1}{n} \frac{dz}{\left((1-z)^{1/n}\right)^{n-1}}$$

x	0	1
z	1	0

$\frac{m-n+1}{n}$

$$\int_1^0 (1-z)^{m/n} (z)^p \cdot -\frac{1}{n} \frac{dz}{(1-z)^{-1/n+1}}$$

$$-\frac{1}{n} \int_1^0 z^p \cdot \frac{(1-z)^{m/n}}{(1-z)^{-1/n+1}} dz$$

$$\frac{1}{n} \int_0^1 z^p \cdot (1-z)^{\frac{m-n+1}{n}} dz$$

$$\frac{1}{n} \beta\left[(p+1), \left(\frac{m-n+1}{n} + 1\right)\right] \Rightarrow \frac{1}{n}\left[\beta(p+1), \frac{m+1}{n}\right]$$

$$\frac{1}{n} \frac{\Gamma(p+1)\, \Gamma\left(\frac{m+1}{n}\right)}{\Gamma\left(p+1+\frac{m+1}{n}\right)}$$

15/2022. $\int_0^1 x^5 (1-x^3)^{10} dx$

let $z = 1-x^3 \Rightarrow x = (1-z)^{1/3}$

$dz = -3x^2 dx$

$dx = -\frac{dz}{3x^2}$

$dx = -\frac{dz}{3(1-z)^{2/3}}$

$\frac{5}{3} \cdot \frac{2}{3}$

$\frac{9}{2}$	0	1
	1	0

$$\int_1^0 (1-z)^{5/3} z^{10} \cdot \left(\frac{-dz}{3(1-z)^{2/3}}\right)$$

$$-\frac{1}{3}\int_1^0 z^{10} \frac{(1-z)^{5/3}}{(1-z)^{2/3}} dz$$

$$\frac{1}{3}\int_0^1 z^{10} \cdot (1-z) \, dz$$

$$\frac{1}{3} \beta(11, 2)$$

Q.16. Show that

$$\int_{-1}^{1} (1+x)^{p-1} (1-x)^{q-1} dx =$$

$$2^{p+q-1} \beta(p,q)$$

Let $2z = 1 + x$, $x = 2z - 1$

$2\,dz = dx$ $\quad 1 - x = 1 - 2z + 1$

$= 2 - 2 - z$

$= 2(1 - z)$

x	0	1
z	0	1

$$\int_0^1 (2z)^{p-1} (2(1-z))^{q-1} \cdot 2\,dz$$

$$2\int_0^1 2^{p-1} (z)^{p-1} \times 2^{q-1} (1-z)^{q-1}\,dz$$

$$2^{p-1+q-1+1} \int_0^1 z^{p-1} (1-z)^{q-1}\,dz$$

$2^{p+q-1}\,\beta(p,q$ formula.